Raft and Fault-Tolerant Key/Value Storage: Consensus and State Machine Replication

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***Abstract*—[TODO: 150–200 words: problem, algorithms (leader election and consensus/replication), containerized environment, metrics, and the main result (linearizable under faults; latency cost under loss/delay).]**

I. Introduction

Replicating a service across several machines is the standard way to make systems resilient to machine crashes, but replication alone is not enough: the replicas must agree on a single order of operations, and they must keep agreeing while links drop messages, nodes leave and reenter the grid, leaders disappear, among other failures. Many consensus algorithms that provide a solution to this problem exist today, and maintaining a log of agreed-upon state-machine operations is a common implementation approach. The Raft algorithm [1] is a widely used consensus protocol that is designed to be understandable and easy to teach, and it is the basis of many production systems, including etcd, Consul, and CockroachDB **[Everton:** cite]. The practical question, however, is not only whether a protocol can be implemented correctly, but what that correctness costs in throughput, tail latency, and recovery time when faults actually happen.

With this in mind, this project had two main elements: 1. Implementing Raft, as well as a simple key-value store system on top of it, with the goal of understanding in detail how this consensus algorithm works; and 2. Building an experimental harness on top of these implementations to measure the cost of fault tolerance in a realistic environment. By implementing the resulting system, key distributed systems concepts were explored, such as leader election, consensus, replication and consistency, fault detection and tolerance, peer-to-peer communication, among others.

The whole system runs as independent OCI containers orchestrated by Docker Compose, with real TCP communication between nodes and an HTTP/JSON interface for clients. **[Everton:** Find a better place for this paragraph]

Basis and contributions

The starting point for this project was the framework supplied by MIT's 6.824 (2020) course labs 2 and 3 [2]. It provided stub APIs, an in-memory simulated network (`labrpc`), a gob (Go's serialization library) wrapper (`labgob`), the Porcupine linearizability checker [3], and, crucially, a comprehensive and thorough set of tests. This skeleton was kept intact and was used to guide and test our implementation; the protocol and service implementations, the runtime transport, persistence, container packaging, observability, and the whole experimental apparatus were written by us. Concretely, this project's contributions are:

- **In-House Raft Implementation:** following MIT 6.824's labs 2A–2C + 3B, our custom raft implementation closely follows Raft's extended paper guidelines, including leader election with randomized timeouts, the log up-to-date voting rule, log replication with fast backup, safety rules for commitment across terms, persistent state restored after a crash, and log compaction with snapshots.
- **Fault-tolerant Key/Value Service Implementation:** following MIT 6.824's labs 3A–3B, a replicated key-value store service was built on top of the Raft implementation built before. It includes client-side leader discovery with retry, response caching, and state snapshotting.
- **Deployment Setup**: the systems' transport layer, ports, and persistence behavior were made configurable via environment variables, so that the same code can run both in the labs' test suites (over the simulated `labrpc` network) and in a containerized deployment (over real `net/rpc`/TCP). A Dockerfile for container image configuration, and a `compose.yaml` file for setting up multiple nodes were provided.
- **Observability and experimentation**: an HTTP API was exposed so that clients can interface with the KV application. A `/status` endpoint was also created, exposing term, leader, commit index, last log index, applied index and /snapshot sizes, plus a benchmark client and a scenario-driven experiment driver (normal operation, network delay/loss, leader crash, partition, cluster-size sweep) that record raw CSVs and check every history for linearizability with Porcupine.

[TODO: Check that this contribution list matches the emphasis of the abstract and of the five-minute video, and trim it if the six-page limit becomes tight.]

The experiments show that the service remains linearizable in every scenario we exercised, including 2% packet loss, leader crashes and partitions, while making the cost of fault tolerance explicit: throughput drops by about two orders of magnitude under 50 ms delay with 2% loss (**[Everton:** this seems awfully high, verigy]), failover takes roughly the election timeout, and the client-visible recovery time is dominated by the client's own RPC deadline rather than by the protocol.

[TODO: Confirm the numbers in the previous paragraph once the final experimental batch is frozen, and keep them

consistent with Table II and Table III.]

The remainder of this report is organized as follows. Section II states the system model and the failure assumptions. Section III describes the architecture and the container deployment. Section IV details the two algorithms and lists the reused code and our modifications. Section V argues the correctness properties and describes the automated tests. Section VI defines the experimental methodology and its reproducibility, Section VII presents the results, Section VIII discusses limitations and threats to validity, and Section IX concludes.

II. System Model

[TODO: State and justify each item below (explicit requirement in the assignment). Do not leave it as a bare list: justify the choices.]

A. Timing model

[TODO: Partially synchronous: no known bound on delay in steady state, but timeouts (500–700 ms election, 150 ms heartbeat) eventually hold, which is what provides liveness. Justify why not purely asynchronous.]

B. Channels

[TODO: Point-to-point channels over TCP (`net/rpc`), no ordering between concurrent requests, with loss/delay injected by `tc netem`; duplication and reordering are handled by idempotence (per-client `ClientId`,`Seq` dedup) and idempotent RPCs. In the test environment, `labrpc` simulates loss, delay and partitions.]

C. Failures

[TODO: Failure model: crash-stop and crash-recovery (state persisted on disk through `Persister`); network partitions; no Byzantine faults. Justify why persisted log plus snapshots suffice for recovery.]

D. Clocks, persistence and restart

[TODO: No synchronized clocks; timeouts use the local clock only as a heuristic. Persistence: `currentTerm`, `votedFor`, log and `lastIncludedIndex` in `raft.state`; snapshot in `kv.snap` (atomic write via temporary file + rename).]

E. Correctness properties

[TODO: Safety: at most one leader per term; Log Matching; if an entry is applied on one node it is applied at the same index on every node; KV linearizability. Liveness: after the network stabilizes and with a live majority, a leader is eventually elected and submitted commands are eventually committed.]

III. Architecture

[TODO: Figure/diagram: HTTP client → nodes (API :9000) and Raft mesh (:8000) between containers; persistent volumes.] **[TODO:** Describe `compose.yaml`: 3+ nodes, bridge network, ports 9000–9002 published, 8000 internal, one volume per node.] **[TODO:** Communication: binary RPC (`net/rpc`/gob) between peers; HTTP/JSON API for clients; `Clerk` on the client with retry and leader switching. Explain why the client (not the server) finds the leader.] **[TODO:** Observability: `/status` endpoint (term, leader, indexes, state and snapshot sizes) used by the experiments.] **[TODO:** Versions: Go, Docker/Compose, base image — fill in Table I.]

IV. Implemented Algorithms

[TODO: Make explicit that these are the two complementary algorithms: leader election (theme 1) and consensus/replication (theme 2).]

A. Leader election and terms

[TODO: Randomized timeouts, voting (log up-to-date rule, one vote per term), conversion to follower on a higher term; role of the observed `leaderId`.]

B. Log replication and consensus

[TODO: `AppendEntries` with `PrevLogIndex/Term`, retry with `XTerm/XIndex/XLen`, `commitIndex` advanced only for entries from the current term (Figure 8 of the paper), snapshots and `InstallSnapshot`.]

C. Linearizable KV state machine

[TODO: Every operation (reads included) goes through the log; response cache keyed by `(ClientId, Seq)` for exactly-once application; replies matched by log index; snapshot of the map plus the cache.]

D. Reused code and modifications

[TODO: Mandatory citation: MIT 6.824 labs 2A–3B skeleton (`labrpc`, `labgob`, `porcupine`, tests) [2]; describe what was changed: the Raft/KV implementation, the `RPCClient` interface for real TCP, the on-disk `Persister`, the `/status` endpoint, the benchmark client and the experiment scripts.]

V. Correctness Properties and Tests

A. Safety arguments

[TODO: Sketch the proof: leader uniqueness per term; majority intersection; the election restriction guarantees the elected leader holds every committed entry; in-order application by index.]

B. Automated tests

[TODO: Describe the inherited suite (2A/2B/2C, 3A/3B) and our own tests: real TCP cluster, follower rejecting writes (HTTP 421), `/status` consistency. Report that everything passes (total time).] **[TODO:** Linearizability: Porcupine reconstructs a valid order for the history; 46/46 runs `linearizable=ok` in the experimental scenarios.]

Table I
EXECUTION ENVIRONMENT

Item	Value
Hardware	**[TODO:** CPU, cores, RAM]
Operating system	**[TODO:** distribution, kernel]
Docker / Compose	**[TODO:** versions]
Go	**[TODO:** version]
Nodes	3, 5, 7 containers (same host)
Load	8 clients, 50 ops/client
Seeds	5–10 per configuration
Warm-up	1 discarded run (sweep)

Table II
THROUGHPUT AND LATENCY (MEDIAN OVER SEEDS)

Scenario	ops/s	p50	p95	p99
Normal	1633	4.5	7.4	8.8
`netem` 50 ms/2%	41	155	307	418
Leader crash (2 nodes)	2038	2.7	8.0	21.9
Partition (after heal)	1227	2.6	6.4	184

VI. EXPERIMENTAL METHODOLOGY

[TODO: Describe the workload (50/50 read/write mix, 5 keys, `kvbench`), the per-run key namespace, and the fact that throughput is measured before the linearizability check.] **[TODO:** Metrics: throughput (ops/s), latency p50/p95/p99, time-to-elect, service recovery time (TTR-service), replica recovery time (TTR-replica), attempts per operation and `ErrWrongLeader` rate. Justify each one.] **[TODO:** Parameters varied at ≥ 3 levels: $N \in \{3, 5, 7\}$; delay/loss via `tc netem` (0, 50 ms, 100 ms; 0%, 2%, 5%); number of clients.] **[TODO:** Fault scenarios: leader crash, leader partition, full cluster restart.] **[TODO:** Reproducibility: `scripts/experiment.py` with recorded seeds, raw CSVs in `scripts/experiments/`, analysis in `scripts/analysis.ipynb`. Exact commands.]

VII. RESULTS

[TODO: Replace the values below with the final version and generate the figures with `scripts/analysis.ipynb`.]

A. Normal operation

[TODO: Describe throughput and percentiles; explain that the 2% retry in the baseline is one extra attempt per client to discover the leader (8/400), not steady-state retrying.]

B. Delay and loss

[TODO: Discussion: throughput drop and p99 increase; why the ceiling is dominated by the election timeout and retransmissions; linearizability is preserved.]

C. Leader crash

[TODO: Time-to-elect $\approx$ election timeout; TTR-service dominated by the client HTTP timeout ($\sim$2 s); TTR-replica measures recovery via log/snapshot.]

Table III
RECOVERY TIMES (MEAN $\pm$ STD)

Scenario	election	TTR-service	TTR-replica
Leader crash	507 ± 114 ms	1868 ± 32 ms	468 ± 124 ms
Partition	349 ± 185 ms	1632 ± 404 ms	not applicable

D. Partition and the disruptive-server problem

[TODO: Inflated terms, churn, attempts per operation; explain that without PreVote the isolated node exiles itself; treat it as a limitation and future work.]

E. Scalability

[TODO: Results with $N = 3, 5, 7$: throughput and latency; discuss why the leader serializes and where replication cost would show up (message counts).]

VIII. LIMITATIONS AND THREATS TO VALIDITY

[TODO: Containers on one host are not independent failure domains; variability from short runs; no batching; no PreVote; no message/byte counters; Docker Desktop environment.]

IX. CONCLUSION AND FUTURE WORK

[TODO: Recap: two algorithms, correctness under faults, performance cost. Future work: PreVote/check-quorum, batching, lease-based reads, incremental snapshots, message accounting, execution across multiple machines.]

REFERENCES

[1] D. Ongaro and J. Ousterhout, "In search of an understandable consensus algorithm," in *2014 USENIX Annual Technical Conference (USENIX ATC 14)*. Philadelphia, PA: USENIX Association, 2014, pp. 305–319. [Online]. Available: https://www.usenix.org/conference/atc14/technical-sessions/presentation/ongaro

[2] R. Morris, F. Kaashoek *et al.*, "6.824 distributed systems: Labs 2 and 3," MIT Computer Science and Artificial Intelligence Laboratory, 2020, skeleton for `labrpc`, `labgob`, `porcupine` and the test suite reused in this work. [Online]. Available: https://pdos.csail.mit.edu/6.824/2020

[3] A. Athalye, "Porcupine: A fast linearizability checker for go," GitHub repository, 2017. [Online]. Available: https://github.com/anishathalye/porcupine